

HISTOLOGY

—————1. With regard to the connective tissue, which statement is false?

- a) Collagen is characterized by high tensile strength and flexibility.
- b) Plasma cells characteristically synthesize and secrete immunoglobulins.
- c) In ordinary connective tissue, the predominant component of ground substance is water.
- d) Fibroblast produces and maintains all extracellular matrix components of connective tissue.
- e) None of the above.

—————2. Concerning the connective tissue, which statement is false?

- a) Proteoglycans are responsible for the highly viscous character of the ground substance.
- b) In blood, the ground substance lacks stabilizing macromolecules.
- c) Reticular fibers are particularly abundant in the framework of hematopoietic organs.
- d) Macrophages are also antigen-presenting cells.
- e) None of the above.

—————3. Which connective tissue cell type is functionally related to the basophils?

- a) Fibroblast
- b) Mast cell
- c) Pericyte
- d) Plasma cell
- e) Myofibroblast

—————4. With regard to cartilage, which statement is false?

- a) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- b) Hyaline cartilage is characterized by an amorphous glassy matrix.
- c) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- d) Fibrocartilage is used for endochondral bone ossification.
- e) Fibrocartilage is present in regions subjected to high tension.

—————5. Elastic cartilage,

- a) is found in the annulus fibrosus and intervertebral disks.
- b) the collagen in its matrix is mainly type II.
- c) is the primary skeletal tissue in the fetus.
- d) has no identifiable perichondrium.
- e) is the most widely distributed cartilage type in the body.

—————6. With regard to bone, which statement is true?

- a) Bone formation without a pre-existing cartilage model is called enchondral bone formation.
- b) Interstitial lamellae represent smooth layers of bone on the surfaces of joints.
- c) Compact (dense) bone is characterized by the presence of osteons.
- d) Chondroblasts secrete organic components of bone matrix.
- e) Osteoblasts are involved in bone resorption.

—————7. Which of the following statements is **correct** with regard to the **bone morphology**?

- a) Primary bone has a higher mineral content than the secondary bone.
- b) Howship's lacunae are calcified collagen bundles of the periosteum that run in concentric manner.
- c) Osteoclasts have processes that run in the form of canaliculi.
- d) Osteoblasts are derived from blood monocytes.
- e) None of the above.

—————8. Which of the following cells **does not** belong to the **mononucleated phagocytic system**?

- a) Microglia
- b) Hepatocytes
- c) Dust cells
- d) Langerhans's cells
- e) Kupffer cells

—————9. With regard to the **white blood cells**, which of the following statements is **true**?

- a) Neutrophils are the most common white blood cells in circulating blood.
- b) A patient with a parasitic infection is likely to reveal an increase in the circulating monocytes.
- c) Basophils normally re-enter the bloodstream after leaving circulation and immigrating into peripheral tissues.
- d) A macrophage can originate from a mast cell.
- e) None of the above.

—————10. Fibrocartilage is not found in the:

- a) Intervertebral discs
- b) Pubic symphysis
- c) Discs of knee joint
- d) Temporomandibular joint
- e) Epiglottis

—————11. Select the ***incorrect statement*** about ***mature erythrocytes***.

- a) do not contain nuclei.
- b) possess haemoglobin, which fills almost the entire cytoplasm.
- c) are unable to move actively, but they are remarkably elastic and can withstand deformation.
- d) contain the organelle peroxisome as oxygen binding protein.
- e) None of the above.

EMBRYOLOGY

_____ 1. Concerning the **fetal membranes**, which statement is **true**?

- a) The yolk sac helps control the embryo's body temperature.
- b) As the bladder enlarges, the umbilical cord involutes to form urachus.
- c) Blood formation occurs in the wall of allantois during the third to fifth weeks.
- d) Primordial germ cells appear in the endodermal lining of the wall of the allantois.
- e) During the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.

_____ 2. Concerning **multiple pregnancies**, which statement is **false**?

- a) The frequency of monozygotic twinning shows marked racial differences.
- b) The rate of dizygotic twinning increases with maternal age.
- c) Dizygotic twins always have two amnions and two chorions.
- d) Dizygotic twinning shows a hereditary tendency.
- e) None of the above.

_____ 3. ***Diethylstilbestrol (DES)*** was once prescribed to women to prevent miscarriages. However, it has since been found to be a teratogen that causes what type of defects?

- a) Reproductive system
- b) Central nervous system
- c) Gastrointestinal
- d) Limb (partial or total absence)
- e) Heart and neural tube defects

_____ 4. Most **birth defects** occur during:

- a) The first 2 weeks
- b) The embryonic period
- c) The fetal period
- d) Early childhood
- e) None of the above

_____ 5. With regard to **birth defects**, which statement is **false**?

- a) Clubfoot, which is due to mechanical forces in the womb, is an example of deformation.
- b) Trisomy of the autosomes occurs with increasing frequency as maternal age increases.
- c) Deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
- d) The phenotype of Turner syndrome is female.
- e) Thalidomide induces limb defects.

_____ 6. Which of the following is **not the function of the amniotic fluid**?

- a) Acts as a barrier to infection
- b) Permits normal fetal lung development
- c) Prevents adherence of the amnion to the embryo and fetus
- d) Cushions the embryo and fetus against injuries by distributing impacts the mother receives
- e) None of the above.

